

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

Kinetic Theory of Gases

Q1

Three moles of ideal gas A with $\frac{C_P}{C_V} = \frac{4}{3}$ is mixed with two moles of another ideal gas B with $\frac{C_P}{C_V} = \frac{5}{3}$.

The $\frac{C_P}{C_V}$ of mixture is (Assuming temperature is constant)

आदर्श गैस A के 3 मोल जिसके लिए कि $\frac{C_P}{C_V} = \frac{4}{3}$ है, को दूसरी आदर्श गैस B के 2 मोल जिसके लिए $\frac{C_P}{C_V} = \frac{5}{3}$ है, के

साथ मिलाया जाता है तो मिश्रण की $\frac{C_P}{C_V}$ का मान बताइए (तापमान को एक-समान मान मानिये।)

(1) 1.5

(2) 1.42

(3) 1.7

(4) 1.3

7th Jan Morning

Sol

(2)

$$\gamma_{\text{mixture}} = \frac{n_1 C_{P1} + n_2 C_{P2}}{n_1 C_{V1} + n_2 C_{V2}} = \frac{n_1 \frac{\gamma_1 R}{\gamma_1 - 1} + n_2 \frac{\gamma_2 R}{\gamma_2 - 1}}{\frac{n_1 R}{\gamma_1 - 1} + \frac{n_2 R}{\gamma_2 - 1}}$$

on rearranging we get पुन व्यवस्थित करने पर,

$$\frac{n_1 + n_2}{\gamma_{\text{mix}} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}; \quad \frac{5}{\gamma_{\text{mix}} - 1} = \frac{3}{1/3} + \frac{2}{2/3}$$

$$\frac{5}{\gamma_{\text{mix}} - 1} = 9 + 3 = 12 \quad \Rightarrow \gamma_{\text{mixture}} = \frac{17}{12} = 1 + \frac{5}{12} \quad ; \quad \gamma_{\text{mix}} = 1.42$$

Q2

In an adiabatic process, volume is doubled then find the ratio of final average relaxation time & initial relaxation time. Given $\frac{C_P}{C_V} = \gamma$

एक रुद्धोष्म प्रक्रम में आयतन को दोगुना कर दिया जाता है, तो अन्तिम तथा प्रारम्भिक औसत विश्रांति काल का अनुपात ज्ञात करें। दिया है $\frac{C_P}{C_V} = \gamma$

(1) $\frac{1}{2}$

(2) 2

(3) $\left(\frac{1}{2}\right)^\gamma$ (4) $\left(\frac{1}{2}\right)^{\frac{\gamma}{2}+1}$

Sol

(Bonus)

relaxation time विश्रांति काल (τ) $\propto \frac{V}{\sqrt{T}}$

and तथा $T \propto \frac{1}{V^{\gamma-1}}$

$$\tau \propto V^{\frac{1+\gamma-1}{2}}$$

$$\tau \propto V^{\frac{1+\gamma}{2}}$$

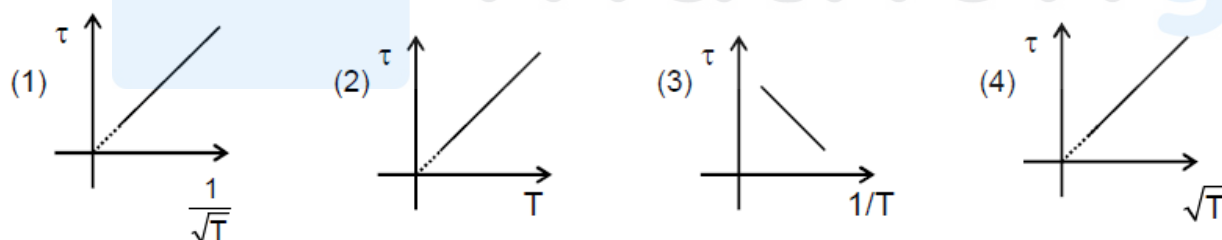
$$\frac{\tau_f}{\tau_i} = \left(\frac{2V}{V} \right)^{\frac{1+\gamma}{2}}$$

$$\frac{\tau_f}{\tau_i} = (2)^{\frac{1+\gamma}{2}}$$

Q3

Which graph correctly represents variation between relaxation time (τ) of gas molecules with absolute temperature (T) of an ideal gas.

इनमें से कौनसा ग्राफ आदर्श गैस अणु के विश्रांती काल (τ) और परम तापमान T के मध्य सही है।

8th Jan Morning

Sol

(1)

$$\tau \propto \frac{1}{\sqrt{T}}$$

Q4

n mole of He and $2n$ mole of O_2 is mixed in a container. Then $\left(\frac{C_P}{C_V}\right)_{\text{mix}}$ will be

He के n मोल तथा O_2 के $2n$ मोल एक पात्र में मिश्रित किये जाते हैं तो $\left(\frac{C_P}{C_V}\right)_{\text{mix}}$ ज्ञात करो।

(1) $\frac{19}{13}$

(2) $\frac{40}{27}$

(3) $\frac{1}{3}$

(4) $\frac{1}{4}$

8th Jan Evening

Sol

(1)

$$\begin{aligned}\gamma_{\text{mix}} &= \frac{n_1 c_{p1} + n_2 c_{p2}}{n_1 c_{v1} + n_2 c_{v2}} \\ &= \frac{n\left(\frac{5}{2}R\right) + 2n\left(\frac{7}{2}R\right)}{n\left(\frac{3}{2}R\right) + 2n\left(\frac{5}{2}R\right)} \\ &= \frac{5+14}{3+10} = \frac{19}{13}\end{aligned}$$

Q5

Two ideal di-atomic gases A and B. A is rigid, B has an extra degree of freedom due to vibration. Mass of A is m and mass of B is $\frac{m}{4}$. The ratio of molar specific heat of A to B at constant volume is :

दो द्विपरमाणु आदर्श गैस A तथा B है। A दृढ़ है। तथा B कम्पन की वजह से अतिरिक्त स्वतंत्रता की कोटि रखती है। A

का द्रव्यमान m और B का द्रव्यमान $\frac{m}{4}$ है। तब नियत आयतन पर A का B से मोलर विशिष्ट ऊष्मा का अनुपात होगा।

(1) $\frac{7}{9}$

(2) $\frac{5}{9}$

(3) $\frac{5}{11}$

(4) $\frac{5}{7}$

9th Jan Morning

Sol

(4)

$$\text{Molar heat capacity of A at constant volume} = \frac{5R}{2}$$

$$\text{A की मोलर विशिष्ट ऊष्मा धारिता नियत आयतन पर} = \frac{5R}{2}$$

$$\text{Molar heat capacity of B at constant volume} = \frac{7R}{2}$$

$$\text{B की मोलर विशिष्ट ऊष्मा धारिता नियत आयतन पर} = \frac{7R}{2}$$

$$\text{Dividing both भाग देने पर, } \frac{(C_v)_A}{(C_v)_B} = \frac{5}{7}$$

Q6

Two gases Ar (40) and Xe (131) at same temperature have same number density. Their diameters are 0.07 nm and 0.10 nm respectively. Find the ratio of their mean free time

समान तापमान पर दो गैसें Ar (40) तथा Xe (131) का संख्या घनत्व समान है। इनके व्यास क्रमशः 0.07 nm तथा 0.10 nm हैं, तो इनके माध्य मुक्त काल का अनुपात ज्ञात करो—

(1) 1.03

(2) 2.04

(3) 3.04

(4) 2.40

Sol

(2)

$$\text{Mean free time} = \frac{1}{\sqrt{2}n\pi d^2 v_{\text{rms}}}$$

$$\frac{t_{\text{Ar}}}{t_{\text{Xe}}} = \frac{d_{\text{Xe}}^2}{d_{\text{Ar}}^2}$$

$$= \left(\frac{0.1}{0.07} \right)^2$$

$$= \left(\frac{10}{7} \right)^2 = 2.04$$

9th Jan Evening

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